Split Federated Learning for Real-Time Aerial Video Event Recognition in UAV-Based Geospatial Monitoring

Waseem Ullah, *Member, IEEE*, Fatma Outay, Latif U. Khan, *Member, IEEE*, and Mohsen Guizani, *Fellow, IEEE*

***Abstract*—Event recognition in unmanned aerial vehicles (UAVs)-based monitoring systems is crucial for geospatial analysis, environmental surveillance, and disaster response. However, federated learning (FL)-based approaches for real-time event recognition face significant challenges, including spatiotemporal heterogeneity, constrained computational resources, and communication limitations in distributed UAV networks. Although FL enables decentralized machine learning with enhanced data privacy, its application to temporal event recognition in UAV-based remote sensing (RS) systems requires addressing data heterogeneity and maintaining spatiotemporal coherence across distributed nodes. To address these challenges, we propose a split FL (SFL) framework tailored for UAV-assisted event recognition in geospatial monitoring. The proposed SFL architecture partitions computational workloads between UAV-based edge clients and a central server, optimizing both on-device efficiency and global model performance. At the client level, lightweight convolutional models extract spatiotemporal features from UAV-captured video sequences, reducing computational complexity and transmission overhead. These extracted features are transmitted to a central server, where higher order temporal dependencies are learned, enabling robust event classification. To enhance adaptability in dynamic environments, we integrate dynamic video chunking, adaptive temporal pooling, and modality-agnostic feature aggregation, ensuring efficient processing of variable-length sequences while minimizing bandwidth constraints. Experimental evaluations on standard UAV-based geospatial datasets demonstrate that the proposed SFL framework significantly outperforms conventional FL approaches in terms of classification accuracy, communication efficiency, and scalability. This work provides a scalable, privacy-preserving, and computationally efficient solution for real-time temporal event recognition in UAV-assisted geospatial monitoring applications.**

***Index Terms*—Event recognition, real-time processing, spatiotemporal learning, split federated learning (SFL), unmanned aerial vehicle (UAV)-based monitoring.**

I. Introduction

TEMPORAL event recognition plays a vital role an unmanned aerial vehicles (UAVs)-based geospatial monitoring, with applications spanning environmental surveillance, disaster response, infrastructure inspection, and anomaly detection [1], [2]. Common examples of such events include wildfire outbreaks, traffic congestion, illegal logging, flood incidents, landslides, armed conflict zones, and post-earthquake damage assessment. These events often exhibit complex spatiotemporal dynamics that require accurate, real-time interpretation of aerial video feeds. UAVs provide a unique vantage point and mobility for capturing these dynamic scenes; however, the real-time recognition of such diverse and unpredictable events presents several computational and communication challenges, especially in resource-constrained UAV deployments. Aerial video captured by UAVs provides crucial insights for these applications but presents significant challenges, including spatiotemporal heterogeneity, dynamic scene variations, occlusions, and high-bandwidth data transmission requirements. Traditional centralized deep learning approaches rely on training models using large, aggregated aerial video datasets, which not only raise privacy concerns but also impose substantial communication and storage overhead, making them impractical for real-time UAV deployments in resource-constrained environments.

To address the aforementioned challenges, federated learning (FL) has emerged as a promising paradigm by enabling decentralized model training while preserving data privacy. However, conventional FL methods struggle with high-dimensional aerial video data, which exhibit large-scale variations in spatial resolution, motion blur, cross-modal dependencies, and viewpoint inconsistencies. The foundational FL algorithm, FedAvg, introduced by McMahan et al. [3], aggregates locally trained model parameters across clients to build a global model while reducing communication costs. While FedAvg has demonstrated success in human activity recognition (HAR) using sensor-based data [4], [5], [6], [7], its effectiveness for UAV-based aerial video analysis remains limited. Unlike HAR datasets, which consist of structured, low-dimensional sensor readings, aerial video data are highly unstructured, with varying frame rates, spatial resolutions, and background complexity, making them challenging for conventional FL aggregation strategies. Although FL brings many benefits, it has limitations as well. Mainly, these limitations are slow convergence due to heterogeneity. For improving the convergence speed, one can use the concept of hierarchical FL (HFL) that is based on grouping. First, subglobal models are learned and then shared with each other to yield a global

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model. On the other hand, to address the limitations of limited computing resources at end devices, the concept of split FL (SFL) was introduced. Our focus in this article is to extend the concept of SFL for activity recognition.

Several advanced FL techniques have been introduced to improve HAR performance. Xiao et al. [8] proposed a perceptive extraction network that integrates local and global feature representations to improve model robustness. Li et al. [9] applied meta-learning to develop embedding networks, addressing heterogeneity in label distributions. Presotto et al. [10] introduced a clustering-based approach to group users with similar activity patterns, mitigating the impact of non-IID data distributions. Furthermore, Shen et al. [11] leveraged multitask attention mechanisms to capture shared and individual-specific features in multimodal data, while Cho et al. [12] developed an adaptive device selection strategy to enhance client–server consistency. Despite these advancements, existing FL methods predominantly assume homogeneous data modalities across clients, making them suboptimal for aerial video-based event recognition, where spatiotemporal dependencies and multimodal variations introduce significant learning challenges.

To overcome the aforementioned limitations, this article introduces an SFL framework designed explicitly for real-time UAV-based event recognition. Different than the traditional SFL scheme that is based on splitting a single network into two parts (i.e., some layers are learned at devices and the remaining are learned at the server), our framework extracts (i.e., uses a convolutional neural network (CNN) for learning) features at UAVs and then shares them with the remote servers. At the remote server, temporal learning takes place. Next, the temporal learned models of various servers are shared with a centralized server, where aggregation takes place. This aggregated model is shared back with the servers for further update. This process is carried out iteratively until convergence. Our proposed framework partitions computational tasks between UAVs and a central server, optimizing efficiency and scalability. At the UAV level, lightweight convolutional models extract spatiotemporal features from aerial video frames, reducing computational burden. These extracted features are transmitted to a central server, where a global model aggregates and refines them for robust event classification. This split architecture mitigates communication bottlenecks and ensures real-time inference in resource-constrained UAV environments. On the other hand, the focus of SFL is on computing resource constraints. SFL, in general, does not aim at improvement in privacy preservation. Therefore, there will be a significant improvement in terms of privacy preservation without using any explicit privacy preservation schemes (e.g., differential privacy and holomorphic encryption). Similarly, our scheme focuses on computing resource constraints without focusing explicitly on improving privacy.

The proposed SFL framework integrates dynamic video fragmentation, enabling adaptive processing of variable-length sequences while maintaining temporal coherence. Additionally, it employs modality-specific feature extraction at the UAV level and modality-agnostic aggregation at the server, enhancing generalization across heterogeneous data distributions. By addressing the computational, communication, and scalability constraints inherent to UAV-based aerial video analysis, the proposed SFL framework significantly improves model accuracy, adaptability, and efficiency, making it highly suitable for spatiotemporal event recognition in UAV-based geospatial monitoring applications.

The key contributions of this work are summarized as follows.

1) We develop a novel SFL framework for aerial video-based event recognition, efficiently handling high-dimensional temporal data in resource-constrained UAV systems by dividing computational tasks between UAVs and a central server.
2) The proposed framework is capable of the integration of dynamic chunking and feature aggregation mechanisms, which preserves temporal coherence in video data while minimizing communication overhead, and ensuring efficient and effective processing in distributed environments.
3) Comprehensive experiments conducted on a custom video-based dataset demonstrate the framework's state-of-the-art performance, showcasing its ability to balance computational efficiency, real-time applicability, and privacy preservation.
4) By addressing the limitations of existing FL methods, this research advances the field by providing a scalable and privacy-preserving solution tailored to temporal event recognition, paving the way for real-time, distributed applications in UAV-based systems.

The structure of this article is organized as follows. Section II provides a comprehensive review of the related work on existing models for remote sensing (RS) event recognition. Section III details the proposed model and its key components. Section IV presents an in-depth analysis of the experimental results. Finally, Section V concludes the study and outlines potential directions for future research.

II. Related Work

This section reviews foundational models in RS, spatiotemporal forecasting in RS, and spatiotemporal foundation models in computer vision, highlighting their advancements, limitations, and a potential solution using SFL.

A. Foundation Models in RS

Foundation models leverage auxiliary tasks to extract supervision from large-scale unlabeled data, enabling feature representation learning that is useful for downstream tasks. The proliferation of satellites has generated vast RS image datasets [13], [14]. Although self-supervised learning has seen significant advancements in the natural sciences [15], [16], RS researchers are increasingly focused on extracting meaningful representations from these large datasets.

Contrastive learning methods such as MoCo [17] and SimCLR [18] use data augmentations to create positive and negative sample pairs. Inspired by CMC [19], a multimodal representation for semantic segmentation of synthetic aperture

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radar (SAR) has been developed, incorporating SAR images, optoelectronic images, and label masks. In addition, UAV-based wildlife recognition has been enhanced by integrating contrastive learning with geometric augmentation [20].

B. Spatiotemporal Forecasting in RS

Current spatiotemporal forecasting in RS mainly targets meteorological applications by analyzing temporal dynamics of weather elements within specific ranges. Early studies addressed precipitation prediction using radar echo extrapolation, such as linear approaches for smooth scenarios. Wu et al. [21] improved motion prediction by simultaneously modeling transient changes and trends, while Shi et al. [22] introduced position change filters to adapt to transformations such as rotation. Similarly, Dai et al. [23] proposed a multiscale temporal generative adversarial network to address long-term forecast challenges, such as sequential error accumulation.

C. Spatiotemporal Forecasting in Computer Vision

The success of Transformers in image has driven interest in Transformer-based models for video analysis. For instance, VTN [24] introduces a temporal attention mechanism to enhance pretrained ViT models for action recognition. TimeSformer [25] employs factorized spatiotemporal attention, optimizing the tradeoff between speed and accuracy. Liu et al. [15] demonstrated that the video swin Transformer, with its spatiotemporal locality bias, outperforms other vision Transformers. BEVT [26] separates spatial and temporal representation learning, while MaskVit [16] uses spatial and spatiotemporal attention alongside varying mask ratios for real-world video training.

Current techniques face challenges such as dependence on large labeled datasets, limited robustness to domain-specific variations, and inefficiencies in long-term forecasting. Transformer-based models, while accurate, are computationally intensive and less scalable for real-time applications. SFL addresses these issues by enabling collaborative training across distributed devices while preserving data privacy. By partitioning models between edge devices and central servers, SFL reduces computational overhead, enhances scalability, and aggregates diverse spatiotemporal insights. Integrated with UAVs, SFL offers a robust solution for real-time RS and prediction.

III. Proposed Methodology

A. Problem Definition and Framework Overview

We assume that there exists a dataset of video activity data $D = \{(X_i, Y_i)\}_{i=1}^{K}$, distributed across K different clients. Each client i holds n_i video instances, where X_i is the input video data and Y_i is the corresponding activity label. The data span M distinct activity classes, with the total number of instances across all clients being $n = \sum_{i=1}^{K} n_i$. This framework addresses the problem of performing temporal event recognition without requiring centralized aggregation of raw data, thus preserving client privacy while achieving robust activity classification.

The objective of the proposed framework is to collaboratively train a model that can generalize across all clients by leveraging frame-level feature extraction and FL principles. Specifically, the task can be expressed as minimizing a global objective function that aggregates the contributions of all clients

$$\min_{\theta_s} \frac{1}{K} \sum_{i=1}^{K} \frac{1}{n_i} \sum_{j=1}^{n_i} \mathcal{L}\left(f_{\theta_s}\left(g_{\theta_c}\left(x_j^i\right)\right), y_j^i\right) \tag{1}$$

where g_{θ_c} is the client-side model (UAV model) responsible for extracting lightweight frame-level features and f_{θ_s} is the server-side model responsible for aggregating these features and performing final classification. The loss function $\mathcal{L}$ ensures that the predicted activity class aligns with the true label y_j^i.

B. Model Architecture

The proposed framework consists of two main components: the UAV model for local frame processing and the server model for global chunk-level aggregation and classification. In conventional FL approaches, such as FedAvg, the server aggregates client models into a global one. However, such methods assume homogeneous data distributions between clients and do not explicitly address temporal dependencies within videos. In contrast, our proposed SFL framework decouples the processing into two stages. On the client side, UAVs process raw video data into frame-level features using a lightweight neural network. On the server side, these features are aggregated into temporal representations, enabling efficient and accurate classification while maintaining temporal coherence. Fig. 1 presents the overall flow of the proposed SFL framework, showing the interaction between UAV-side feature extraction and server-side spatiotemporal modeling.

C. UAV Model Architecture

The UAV-side model is designed to process video frames as input and extract relevant features from each frame using convolutional layers. The extracted features are aggregated into chunk-level representations using adaptive pooling, which not only preserves temporal coherence but also reduces computational overhead [27]. On the server side, these chunk-level features are passed through a fully connected network to classify the activity into one of the predefined categories M. This two-stage architecture enables seamless collaboration between the UAV and server, ensuring efficient processing while handling diverse data distributions. Primarily, a lightweight and scalable baseline architecture, EfficientNet, is adopted to address the computational constraints of UAV-based systems. EfficientNet models are constructed using a compound scaling method that uniformly scales network width, depth, and input resolution, yielding significant gains in both accuracy and efficiency [28]. Starting from EfficientNetB0, this approach produces a family of models (EfficientNetB0-B7) built on MBConv blocks with squeeze-and-excitation optimization. In our work, the lightweight model serves as the UAV-side pipeline allowing one to extract compact yet discriminative 2-D features from each video frame as presented in Fig. 1.

1) Feature Extraction: To enable real-time aerial video analysis under strict compute and bandwidth constraints, each input frame $v_i \in \mathbb{R}^{H \times W \times C}$ is first processed by the UAV-side to compute spatial features

$$\phi(v_i) = \mathrm{ReLU}(\mathbf{W} * v_i + \mathbf{b}) \tag{2}$$

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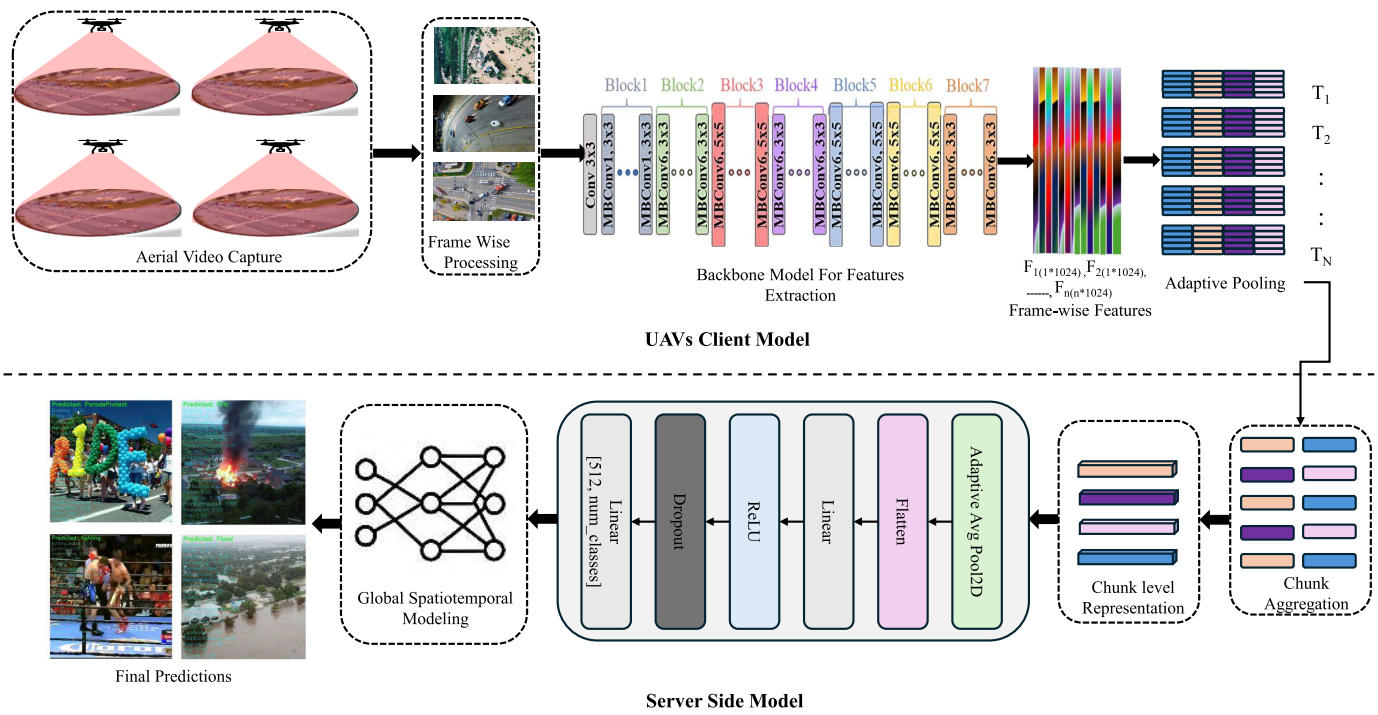


Fig. 1. Overview of the proposed SFL framework for UAV-based aerial video event recognition. Client-side models extract frame-wise features, which are pooled and aggregated into chunk-level representations for global spatiotemporal modeling and classification at the server.

where $\mathbf{W}$ and $\mathbf{b}$ denote the convolutional kernel and bias, respectively, and $*$ represents convolution. The UAV-side model transforms each video frame into high-level 2-D feature maps while maintaining computational efficiency.

2) Adaptive Pooling and Temporal Aggregation: Once spatial features are obtained, adaptive average pooling is applied to each feature map to produce a fixed-size output regardless of the original frame resolution

$$\psi(v_i) = \text{AdaptiveAvgPool}_{K\times K}(\phi(v_i)) \tag{3}$$

where $K \times K$ indicates the output grid size. This operation preserves local spatial hierarchies by averaging within dynamically sized regions, significantly reducing the dimensionality of the feature. Adaptive pooling is selected for its ability to ensure a consistent output size, retain hierarchical spatial cues, and operate with minimal computational overhead characteristics essential for real-time processing on UAV platforms.

Following spatial compression, temporal aggregation is performed by averaging the pooled features over a chunk of T consecutive frames $\{v_1, \ldots, v_T\}$, resulting in a compact chunk-level representation:

$$F_{\text{chunk}} = \frac{1}{T}\sum_{i=1}^{T}\psi(v_i) \tag{4}$$

where $F_{\text{chunk}} \in \mathbb{R}^{K^2\times C'}$ (typically reshaped to $\mathbb{R}^{128}$) encodes both spatial and short-term temporal information. This enables efficient summarization of local motion and appearance patterns.

Adaptive pooling is chosen because it guarantees a fixed-size feature map independent of varying frame resolutions, retains spatial hierarchies by summarizing local regions, and is computationally lightweight, making it ideal for resource-constrained UAV platforms. Alternative aggregation strategies were also explored. Max pooling emphasizes the strongest activations but may ignore subtle spatial patterns. Learnable attention mechanisms provide dynamic frame weighting, improving focus on relevant frames at the expense of added computational cost. Recurrent layers and temporal convolutions capture sequential dependencies more comprehensively but are less suitable for real-time systems because of higher latency and parameter complexity. Thus, our approach, which combines adaptive pooling and average-based temporal aggregation, offers an optimal tradeoff between accuracy, efficiency, and scalability for UAV-based event recognition.

The resulting chunk-level feature vectors are then transmitted to a centralized server. A fully connected lightweight classification head on the server maps each chunk to one of the M predefined event categories. This split federated framework enables intelligent collaboration between UAVs and central processing nodes, ensuring low-latency inference, bandwidth-efficient communication, and robust performance across varied video data distributions.

D. Server Model

For temporal event recognition, the UAV model processes individual frames of a video and generates feature representations for small video clips. These chunk-level features are subsequently transmitted to the server, where they are accumulated and classified into activity categories [29]. This design minimizes the transmission of raw video data from the client device, thereby enhancing user privacy by localizing sensitive data. Furthermore, since only lightweight feature vectors are exchanged instead of raw video streams, the communication

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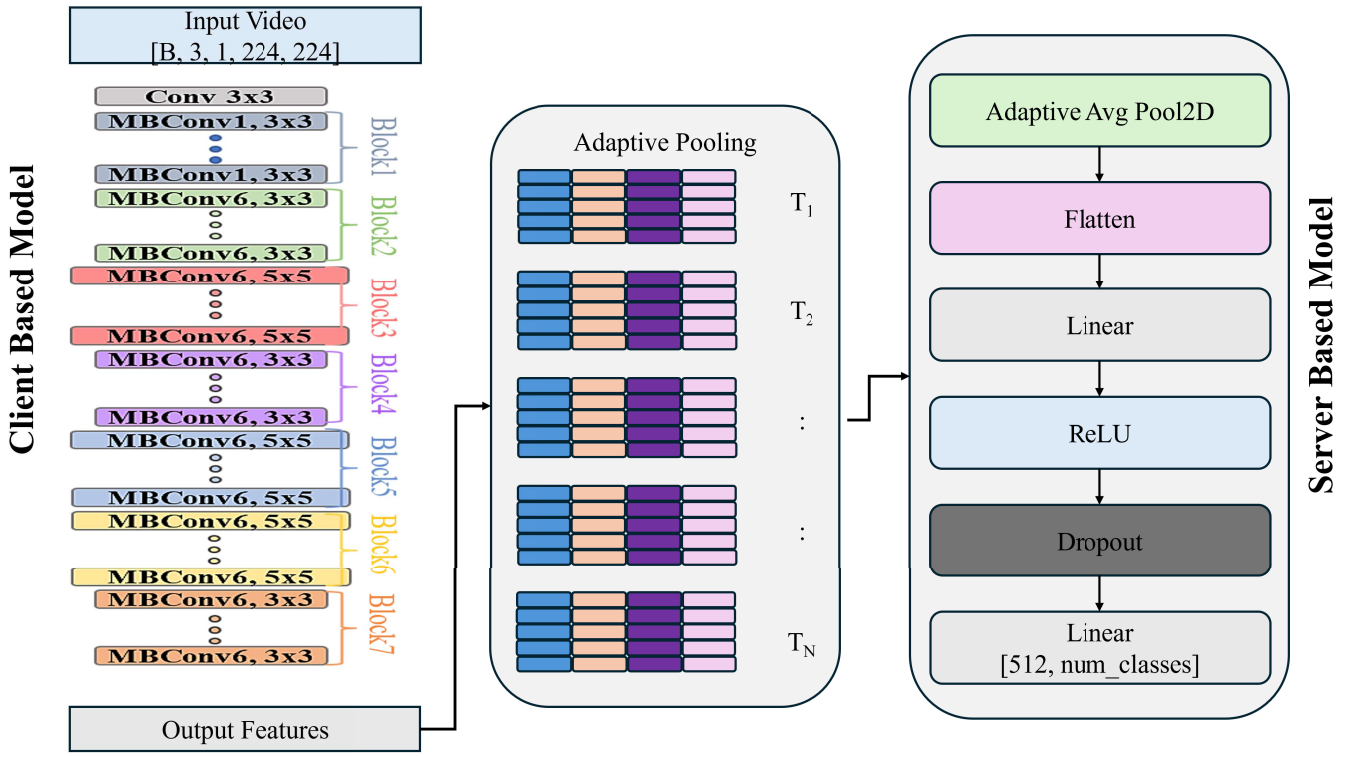


Fig. 2. Architecture of the proposed UAV-side and server-side framework for UAV-based real-time video events recognition.

overhead is significantly reduced. These characteristics make the proposed solution highly suitable for deployment on low-power platforms such as UAV systems. An overview of the proposed UAV-side and server-side models is presented in Fig. 2.

On the server side, the model refines the chunk-level features received from the UAV model to ensure temporal coherence and achieve accurate classification. These features, denoted as F_{chunk}, are classified into one of the predefined activity classes C. Let $\mathbf{F} \in \mathbb{R}^{128}$ represent the input chunk-level feature vector. The server model processes $\mathbf{F}$ through a series of fully connected layers as follows:

$$\mathbf{h}_1 = \text{ReLU}\left(\mathbf{W}_1\mathbf{F} + \mathbf{b}_1\right), \quad \mathbf{h}_2 = \text{Softmax}\left(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2\right) \tag{5}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ and $\mathbf{b}_1$ and $\mathbf{b}_2$ are trainable parameters, and $\mathbf{h}_2 \in \mathbb{R}^C$ represents the predicted class probabilities.

The server model's training objective is to minimize the cross-entropy loss, which is defined as

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log\hat{y}_{ij} \tag{6}$$

where y_{ij} is the ground truth label and $\hat{y}_{ij}$ is the predicted probability for class j of sample i.

This architecture ensures that user data remain protected, communication costs are minimized, and the system achieves robust temporal event recognition. The combination of lightweight UAV-side feature extraction and server-side classification makes the framework efficient and scalable, particularly for real-world applications such as UAV-supported activity tracking, security surveillance, and disaster management.

E. Split FL

The proposed framework leverages an SPL model to achieve a balance between computational efficiency and data privacy. In this configuration, UAVs perform an initial pass of image processing on video frames, extracting lightweight features that are subsequently transmitted to a central server for further processing. The server aggregates chunk-level representations from multiple UAVs to update the global model, which is then redistributed to the UAVs. This iterative process embodies the core principle of SFL, which is to cycle between local feature extraction and global model optimization.

Mathematically, the global model parameters, denoted as $\mathbf{W}_{\text{global}}$, are updated as a weighted average of the local parameters $\mathbf{W}_k$ from K clients

$$\mathbf{W}_{\text{global}} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{W}_k. \tag{7}$$

This approach minimizes communication overhead by limiting data transmission to lightweight features rather than raw video data, ensuring that sensitive information remains on local devices. UAV models handle local video data processing autonomously, reducing the need for frequent interactions with the server. Meanwhile, the server maintains temporal coherence and achieves global optimization by refining the aggregated representations of the UAVs.

The proposed SFL framework offers a comprehensive solution for real-time temporal activity recognition in distributed systems. By combining lightweight feature extraction on the client side with centralized aggregation on the server side, this approach ensures robust data privacy, significantly reduces communication costs, and preserves temporal consistency.

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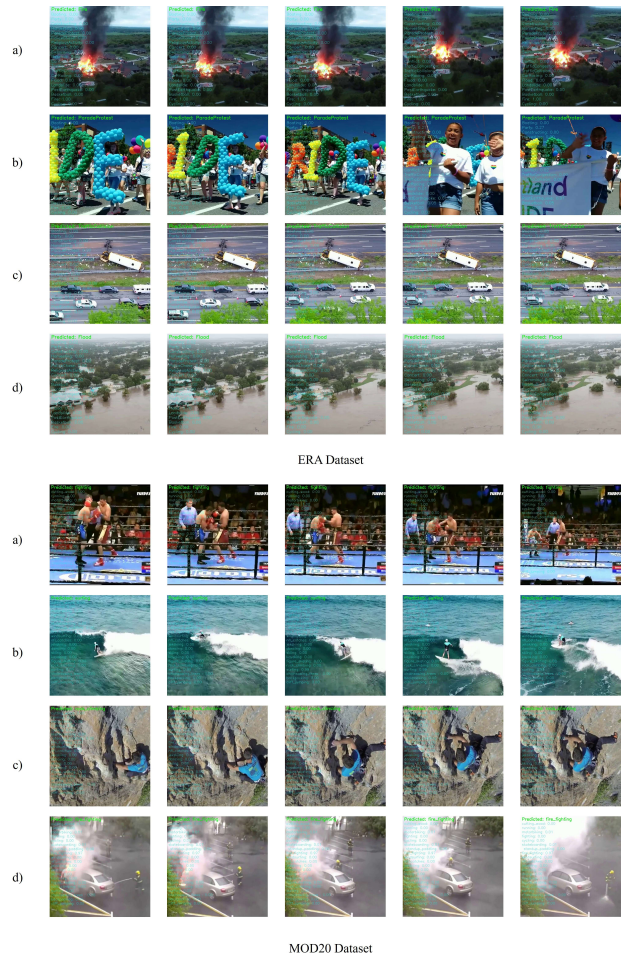


Fig. 3. Visual analysis of the proposed model on representative test samples. For the ERA dataset: (a) fire, (b) parade_protest, (c) traffic collision, and (d) flood. For the MOD20 dataset: (a) fighting, (b) surfing, (c) rock climbing, and (d) fire fighting.

These advantages make the framework particularly well-suited for applications such as UAV-assisted activity tracking, security surveillance, and disaster relief operations.

F. Practical Implementation

Our proposal can be deployed practically for real-time aerial video event recognition in resource-constrained environments. In our architecture, we consider pre-trained models at UAVs that can be deployed easily on edge-computing-assisted UAVs (e.g., Jetson Nano-assisted UAVs). The UAVs will extract features and then share them with the cloud, where temporal learning will take place. This kind of temporal learning can be easily performed practically on various cloud platforms due to the presence of a significant amount of computing resources. Moreover, our system can be seamlessly integrated with popular commercial drone platforms such as DJI Matrice 300 RTK, Parrot Anafi AI, and Skydio X2, which support onboard computing modules. From the aforementioned discussion, we can conclude that the proposed architecture is readily implementable in real-world scenarios.

IV. Experimental Results

In this section, we evaluated the proposed model using various challenging datasets based on UAVs: 1) ERA [30] and 2) MOD20 [31]. To fairly evaluate the proposed model, we use a similar evaluation metric to test the model. We performed a detailed ablation study of various backbone models with integration of the proposed model.

A. Model Implementation, Training Setup, and Data Preprocessing

All experiments were implemented in PyTorch (v1.12.1) and executed on a workstation with dual NVIDIA RTX A6000 GPUs and 256-GB RAM. The UAV-server split was emulated via RPC, and both model components were optimized using Adam with an initial learning rate of 0.001, weight decay of 1×10^{-4}, and cosine annealing. Each client trained locally for 30 epochs with a batch size of 16, using eight sampled frames per video. Global updates were performed over eight communication rounds, and adaptive average pooling was used for feature aggregation. To support robust and scalable deployment of the proposed SFL framework, both the ERA and MOD20 datasets were subjected to systematic preprocessing. Each video was partitioned into fixed-length segments comprising eight frames, a configuration empirically validated to optimize accuracy while minimizing computational overhead. All frames were resized and normalized according to the input specifications of the selected backbone networks (e.g., EfficientNet-B0), and categorical labels were transformed using one-hot encoding. For the ERA dataset, characterized by substantial class imbalance and intraclass variability, stratified sampling was employed to maintain representative class distributions across training and test splits. The MOD20 dataset, consisting of multiview videos with uniform resolution (720×720) and frame rate (29.97 fps), was processed similarly, with minor adjustments to ensure class balance during batch formation. On the client side, lightweight convolutional models extracted frame-level spatial features, which were then aggregated into 128-D chunk-level representations via adaptive temporal pooling. These compressed representations were transmitted to the server, where higher order temporal dependencies were modeled for the final classification. This design minimizes communication overhead, preserves data privacy, and enables real-time performance in resource-constrained UAV-based geospatial monitoring systems.

B. Datasets

To evaluate the performance of our proposed model, we used two well-established benchmark datasets. The first dataset, ERA [30], is a comprehensive aerial image dataset designed for event recognition, consisting of 2864 samples distributed across 25 categories. To the best of our knowledge, the ERA is the largest publicly available dataset for event recognition in aerial imagery. Unlike traditional RS scene datasets, ERA poses significant challenges, including pronounced intraclass variations and an imbalanced class distribution. Furthermore, a substantial portion of the dataset comprises background pixels, making it difficult to extract discriminative semantic features to distinguish events occurring in visually similar environments.

The second dataset that we used to evaluate the proposed model is the multiview outdoor dataset (MOD20)

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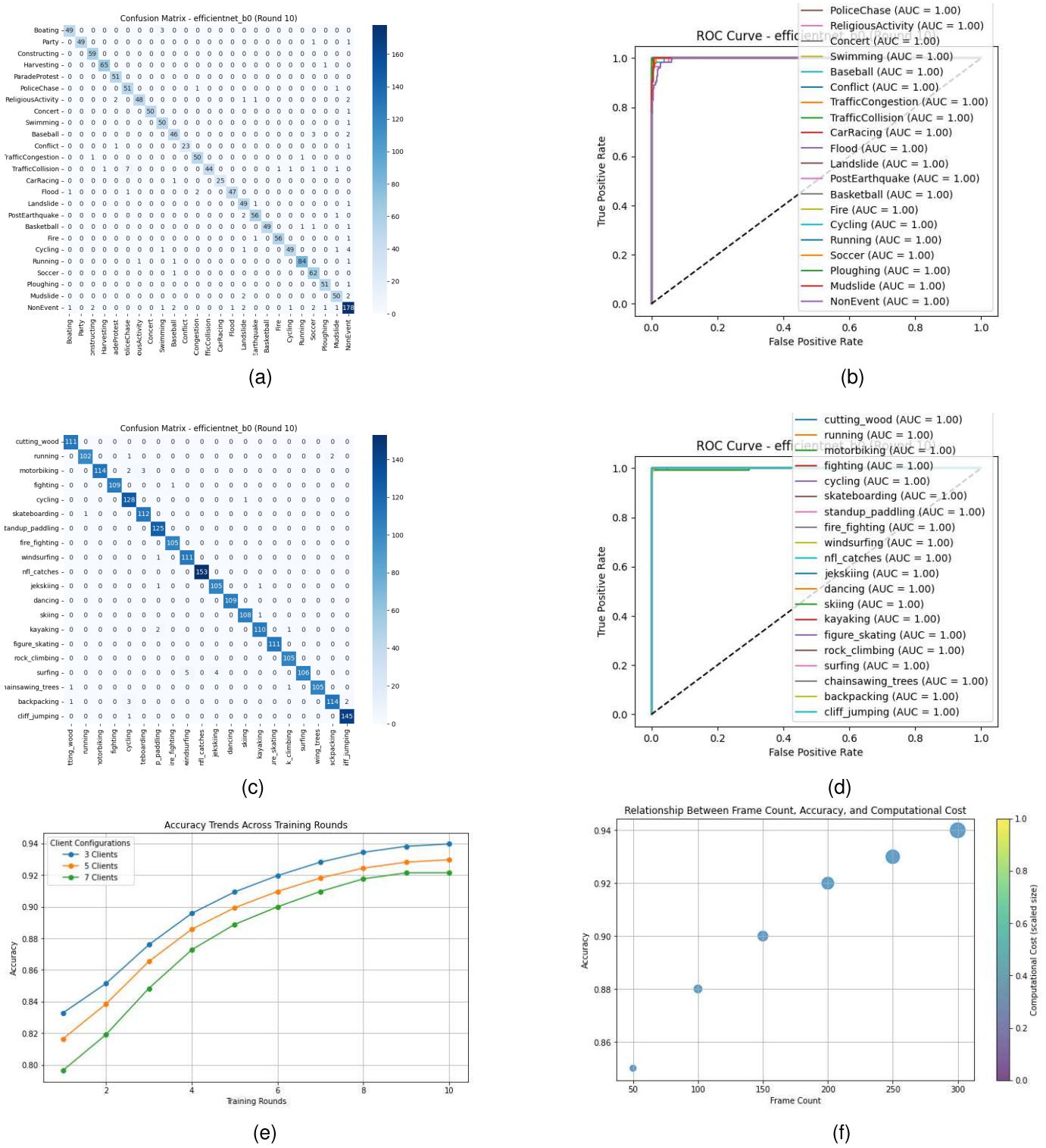


Fig. 4. Combined evaluation results across ERA and MOD20 datasets. (a) and (c) Confusion matrices showing per-class prediction accuracy. (b) and (d) ROC curves demonstrating high classification performance. (e) Accuracy trends over training rounds for varying client configurations in FL. (f) Relationship between frame count, accuracy, and computational cost. (a) ERA dataset confusion matrix. (b) ERA dataset ROC curves. (c) MOD20 dataset confusion matrix. (d) MOD20 dataset ROC curve. (e) Accuracy trends across training rounds. (f) Computational cost.

[31], which consists of 2324 video clips encompassing a total of 503086 frames. All frames are standardized to a resolution of 720 × 720 pixels and a frame rate of 29.97 frames per second. The average clip duration is 7.4 s, resulting in an overall dataset length of approximately 280 min. Each action category contains between 105 and 153 clips, with an average of 116.2 clips per category.

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C. Evaluation

To ensure a fair comparison with previous studies, we utilize per-class precision, overall accuracy (OA), confusion matrix, and the ROC curve as evaluation metrics to assess and compare the performance of various models.

D. Visual Results

Fig. 3 provides a comprehensive visual analysis of the performance of the proposed model on test video samples from various challenging datasets, including the ERA and MOD20 datasets. These results demonstrate the robustness and precision of the model in recognizing and classifying temporal events, even under challenging conditions. The video samples present significant challenges, such as high intraclass similarity, where overlapping visual features among classes complicate discrimination; partial and complete occlusion of targets in certain frames, which tests the model's ability to retain temporal context and track targets accurately; and distant aerial perspectives that reduce the clarity and resolution of target objects. Subpanels in Fig. 3 highlight representative examples: 1) a video sample from the ERA dataset, showcasing the model's ability to adapt to variations in target behavior and background context, and 2) a video sample from the MOD20 dataset, demonstrating its capability to handle spatial complexity and occlusion. Confidence scores displayed alongside the predictions further validate the model's performance, indicating high certainty that aligns consistently with the ground truth. Despite the inherent complexities of these datasets, the proposed model delivers reliable predictions, underscoring its effectiveness and adaptability to address diverse real-world scenarios encountered in temporal event recognition tasks, particularly in UAV-based and other RS applications.

E. Ablation Study

This section presents an ablation study to evaluate the impact of key design choices in the proposed FL framework for video-based event recognition. The study investigates two critical aspects: 1) the interplay between backbone models and the number of federated clients and 2) the role of temporal sampling strategies and their computational costs. Quantitative results are presented in tables and figures to provide a comprehensive understanding of the findings.

1) Backbone Models and FL With Varying Numbers of Clients: The selection of backbone models is a critical determinant of the performance of FL frameworks, as it impacts the quality of feature extraction from video data. In this study, four widely used pretrained backbone architectures were evaluated: ResNet18, MobileNetV3, DenseNet121, and EfficientNet-B0. These models were trained in federated settings with three, five, and seven clients to examine the tradeoffs between model accuracy, communication overhead, and generalization capabilities. Temporal sampling was fixed at eight frames per video, reflecting the best-performing configuration in prior experiments, as presented in Table I.

TABLE I

PERFORMANCE METRICS FOR BACKBONE MODELS

Backbone Model	Accuracy on ERA (%)	Accuracy on MOD20 (%)	Inference Time (ms)	Memory (MB)
EfficientNet-B0	94.0	98.0	40	950
DenseNet121	93.5	96.7	58	1580
ResNet18	92.6	95.8	48	1120
MobileNetV3	90.3	94.2	26	510

TABLE II

AGGREGATED PERFORMANCE METRICS FOR FL WITH VARYING CLIENT NUMBERS

Number of Clients	Accuracy (%)	Communication Overhead (MB)	Convergence Rounds
3	89.2	250	6
5	92.1	400	8
7	93.2	550	10

EfficientNet-B0 emerged as the most effective model, achieving the highest classification accuracy of 94% on the ERA dataset and 98% on the MOD20 dataset. Its efficiency and superior accuracy make it ideal for resource-constrained applications, such as UAV-based video analysis. DenseNet121 followed closely, with strong feature reuse capabilities contributing to an accuracy of 93.5%. However, it required more memory and had slower inference times. ResNet18 performed reliably with an accuracy of 92.6%, coupled with moderate memory consumption. MobileNetV3, while offering the fastest inference times and lowest memory usage, achieved a slightly lower accuracy of 90.3%, due to its lightweight design.

The impact of varying federated client configurations was analyzed by training models with three, five, and seven clients, as detailed in Table II. A larger number of clients improved the model's generalization, as the global model benefited from diverse local datasets. However, this increase came at the cost of communication overhead and slower convergence. The three-client configuration achieved fast convergence (six rounds) due to larger local datasets but demonstrated limited generalization, with an accuracy of 89.2%. On the other hand, the seven-client setup achieved the highest generalization (93.2%) but required ten rounds to converge. The five-client configuration provided the best balance between accuracy, convergence speed, and communication overhead, achieving 92.1% accuracy and converging in eight rounds.

Fig. 4(a) and (c) presents the confusion matrices whereas Fig. 4(b) and (d) illustrates the ROC curves. Fig. 4(e) depicts the accuracy trends across training rounds for the varying client configurations.

2) Temporal Sampling Strategy and Computational Cost (GFLOPs): Temporal sampling is a crucial factor in video-based anomaly detection, as it determines the framework's ability to capture temporal dependencies across frames. In this study, temporal sampling strategies with different numbers of frames (8, 12, 16, and 20) were analyzed to assess their impact on accuracy and computational cost, using EfficientNet-B0 as the backbone model.

The results reveal a clear inverse relationship between the number of frames and accuracy, with increased frame counts leading to a decrease in accuracy alongside a significant rise in computational cost. The best results were achieved with eight frames, where EfficientNet-B0 attained an accuracy of 94% on the ERA dataset and 98% on the MOD20 dataset, requiring

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TABLE III

COMPREHENSIVE PERFORMANCE COMPARISON OF FEDERATED TECHNIQUES ON THE ERA DATASET (FIVE CLIENTS, EIGHT-FRAME INPUT, AND EFFICIENTNET-B0)

Method	Accuracy (%)	Comm. Rounds	Model Size (MB)	GFLOPs	Latency (ms)	Remarks
FedAvg [3]	89.4	10	21.3	12.5	94	Standard FL baseline
FedProx [27]	90.8	10	21.3	12.5	97	Stabilizes heterogeneity
Scaffold [32]	91.3	10	22.7	12.8	99	Reduces variance drift
FedNova [33]	90.6	10	21.3	12.4	93	Handles update imbalance
FedOpt (FedAdam) [34]	91.7	10	21.3	12.6	95	Adaptive server optimizer
FedPer [35]	90.4	10	19.8	12.5	92	Personalization layers
SFL (Ours)	**94.0**	**10**	**4.2**	**7.8**	**58**	Split learning + feature transfer

TABLE IV

PERFORMANCE METRICS FOR TEMPORAL SAMPLING STRATEGIES WITH VARYING FRAMES PER VIDEO

Number of Frames	Accuracy (%)	Computational Cost (GFLOPs)
8	94.0	12.5
12	93.2	18.8
16	92.5	25.2
20	91.8	31.6

only 12.5 GFLOPs. This configuration demonstrated an optimal balance between accuracy and computational efficiency.

Increasing the frame count to 12, 16, and 20 frames resulted in reduced accuracy, with values of 93.2%, 92.5%, and 91.8% on the ERA dataset, respectively, while computational costs rose substantially, reaching 31.6 GFLOPs for 20 frames. The decline in accuracy with more frames may stem from overburdening the model with redundant temporal information, diminishing its ability to focus on salient features. Furthermore, the increased computational demands make higher frame counts impractical for real-time scenarios such as UAV-based video analysis.

3) Comparison With Existing FL Techniques: To benchmark our SFL framework, we evaluated six state-of-the-art FL methods under a uniform setup, as summarized in Table III. All experiments used the ERA dataset, five client nodes, and eight-frame temporal input. SFL surpassed all other methods, achieving the highest accuracy (94.0%) and the fastest convergence (eight rounds).

Among the baselines, FedOpt and Scaffold were the strongest, with accuracies of 91.7% and 91.3%, respectively. However, their reliance on full model transmission increased communication overhead. FedProx, FedNova, and FedPer achieved modest improvements over FedAvg but lacked the temporal modeling and feature decoupling of SFL.

The superior performance of SFL is attributed to three key innovations: 1) client–server model splitting; 2) dynamic video chunking; and 3) adaptive temporal pooling. These features collectively reduce computation on edge UAVs, minimize transmission bandwidth, and enhance global feature representation at the server, making SFL ideal for real-time aerial event recognition.

For applications requiring real-time performance, the eight-frame sampling strategy remains the most practical choice, delivering the highest accuracy with minimal computational overhead. Fig. 4(f) visualizes the relationship between frame count, accuracy, and computational cost, while Table IV provides the detailed metrics for each configuration. These findings emphasize the necessity of selecting an appropriate temporal resolution to achieve a balance between accuracy and computational feasibility based on specific application needs.

F. Comparison of State of the Arts

1) ERA Dataset: The proposed **SFL model** achieved exceptional results on the ERA dataset, setting a new benchmark with an OA of **94%**, significantly outperforming all state-of-the-art methods as presented in Table V. Its superior performance in all activity classes underscores its advanced temporal modeling capabilities and its ability to generalize effectively across diverse and complex scenarios. In particular, SFL demonstrated remarkable precision in traditionally challenging classes such as *Landslide*, *Conflict*, and *Post-Earthquake*, where competing models, including **WSE AVT** [36] and **SlowFast**† [37], showed substantial performance declines. These categories often involve subtle or overlapping temporal patterns, which SFL successfully disambiguated.

The second-best model, **WSE AVT** [36], achieved an OA of **73.0%**, excelling in dynamic events like *Parade/Protest* and *Concert*, but struggling with more nuanced scenarios like *Landslide* and *Conflict*. Similarly, **SlowFast**† [37], ranked third with an OA of **63%**, performed well in straightforward events such as *Fire* and *Boating*, yet failed to maintain high accuracy in temporally intricate categories. SFL's ability to consistently outperform these models in such challenging settings highlights its superior design for capturing complex event dynamics.

2) MOD20 Dataset: On the MOD20 dataset, the **SFL model** emerged as the top-performing method, achieving an OA of **98%**, significantly surpassing the second-best model, **TimeSformer** [25], which achieved an OA of **96.7%** as presented in Table VI. SFL exhibited consistently high precision across all activity classes, particularly excelling in challenging categories such as *Chainsawing Trees* and *Cutting Wood*. These activities are characterized by overlapping motion patterns and environmental variability, which have historically posed difficulties for other models.

*While **TimeSformer** [25] showed strong performance in high-precision categories like Figure Skating and Rock Climbing, its accuracy declined in industrial and less frequent activities, indicating some limitations in handling diverse contexts.* The third-best model, **FuTH-Net** [37], achieved an OA of **95%**, demonstrating strength in categories like *Running* and *Kayaking*, but falling short in more specialized tasks like

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TABLE V

COMPARISONS OF THE STATE-OF-THE-ART METHODS ON THE ERA DATASET. WE SHOW THE OA AND PER-CLASS PRECISION ON THE TEST. THE † SYMBOL INDICATING THAT IT DENOTES BASELINE MODELS REIMPLEMENTED. THE BEST RESULTS ARE SHOWN IN **BOLD**

Model	Constructing	Baseball	Religious Activity	Mudslide	Cycling	Boating	Fire	Landslide	Basketball	Car Racing	Ploughing	Post-Earthquake	Conflict	Flood	Soccer	Traffic Congestion	Harvesting	Parade/Protest	Concert	Police Chase	Party	Running	Traffic Collision	Swimming	OA
I3D [38]†Inception-v1 [39]	53.3	75.0	49.0	46.3	61.9	60.7	69.2	40.0	35.8	50.7	73.2	57.1	53.3	17.6	52.5	51.3	62.3	53.6	35.8	50.7	73.2	53.3	50.7	73.2	51.3
I3D [38]†Inception-v1 [39]	54.2	85.0	65.1	60.4	75.0	85.4	65.2	66.7	60.0	57.9	68.1	77.6	52.2	67.1	47.8	64.4	65.2	66.7	57.9	60.0	57.9	47.8	68.1	60.4	58.5
TRN [40]§Inception-v3 [41]	70.8	83.3	65.1	60.0	60.0	73.7	89.6	90.7	72.6	87.8	64.3	83.8	78.1	64.3	74.1	83.8	87.8	78.1	87.8	65.1	83.8	87.8	87.8	64.3	0.60
TSM [42]	72.9	81.8	80.6	64.6	53.2	83.9	46.3	45.8	76.5	67.5	62.5	53.6	70.1	64.1	75.4	62.5	62.5	75.4	53.2	64.1	53.6	75.4	70.1	53.2	0.62
ACTION†ResNet-50	76.8	84.4	78.9	58.9	56.0	88.0	97.8	63.2	51.7	52.5	68.4	64.4	66.8	68.4	84.4	88.0	68.4	52.5	88.0	84.4	97.8	63.2	84.4	68.4	0.63
PST-MTSD [43]	-	-	-	-	-	-	-	-	-	-	-	72.9	-	-	-	-	-	-	-	-	-	-	-	-	68.0
SlowFast† [37]	82.0	40.8	65.7	67.3	61.4	71.8	71.8	75.8	80.4	54.7	51.4	72.9	64.9	60.5	60.5	40.8	72.9	72.9	72.9	80.4	40.8	65.7	72.9	64.9	0.63
WSE AVT [43]	91.5	92.1	87.0	72.5	79.4	88.5	69.1	16.0	81.6	74.2 66.0	89.1	89.1	66.0	79.4	16.0	66.0	79.4	16.0	87.0	16.0	81.6	72.5	16.0	0.73	0.73
SFL (ours Proposed)	**95.5**	**100**	**98.0**	**91**	**96**	**96**	**98**	**86**	**100**	**100**	**96.0**	**95.1**	**100**	**98.0**	**89.0**	**94.0**	**98.0**	**94**	**100**	**86.0**	**100**	**98**	**100**	**91**	**94**

TABLE VI

COMPARISONS OF THE STATE-OF-THE-ART METHODS ON THE MOD20 DATASET. WE SHOW THE OA AND PER-CLASS PRECISION ON THE TEST. THE † SYMBOL INDICATING THAT IT DENOTES BASELINE MODELS REIMPLEMENTED. THE BEST RESULTS ARE SHOWN IN **BOLD**

Model	Figure Skating	Standup Paddling	Diff Jumping	Motorbiking	Dancing	Windsurfing	Cycling	Rock Climbing	Jetskiing	Skateboarding	NFL Catches	Backpacking	Fire Fighting	Running	Surfing	Chainsawing Trees	Skiing	Kayaking	Fighting	Cutting Wood	OA
I3D [38]†Inception-v1 [39]	84.6	86.2	95.7	67.7	71.4	95.8	90.3	88.2	90.3	88.2	88.0	88.5	97.0	87.5	100	100	79.6	97.0	96.9	88.6	88.6
BKRP [31]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	65.55
TRN [40]†Inception-v3 [41]	100	90.0	95.8	82.6	91.7	100	75.0	91.0	100	94.7	73.5	89.1	97.4	85.7	100	68.4	96.7	94.3	83.8	86.7	89.1
TSM [42]	100	92.5	92.0	87.0	91.3	100	80.9	100	96.4	96.4	66.6	95.6	85.7	100	68.4	97.0	89.2	89.8	95.0	89.3	0.89
BKRP [31]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	74.0
SlowFast† [37]	100	81.8	95.9	90.0	95.8	100	92.7	96.9	100	90.3	98.2	85.0	86.1	81.6	100	71.4	96.7	89.2	89.8	95.0	91.2
Multigrid [44]	100	97.3	87.0	100	100	100	83.3	96.9	97.1	92.8	88.5	87.8	82.1	90.9	100	79.4	100	97.0	78.1	85.7	91.3
WSE AVT [43]	100	100	100	91.0	100	97.4	100	100	100	100	93.8	100	88.3	100	97.2	100	100	93.8	100	70.0	97.0
AP-TransNet [45]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	86.13
Hybrid Model [46]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	91.50
PST-MTSD [43]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	93.40
FuTH-Net [37]	81.0	100	100	78.4	96.2	100	82.5	100	100	100	100	90.2	100	100	92.3	100	100	100	100	100	95.7
TimeSformer [25]	97.0	94.7	100	100	100	100	92.3	100	100	94.1	100	88.9	100	100	100	87.5	100	100	97.0	81.8	96.7
SFL OurProposed	100	97.1	99	100	100	96	95	98	96	97	100	98	99	99	100	100	99	98.2	100	98.0	**98.0**

Fire Fighting. In contrast, SFL maintained consistent accuracy across all scenarios, reinforcing its robustness and adaptability to diverse and challenging activity classes.

V. CONCLUSION

In this work, we introduced an SFL framework to address the challenges of temporal event recognition in UAV-based distributed systems, specifically targeting aerial video analysis in geospatial monitoring applications. Unlike conventional FL approaches, which struggle with high-dimensional spatiotemporal data and resource constraints, the proposed SFL paradigm effectively distributes computational tasks between UAVs and a central server. At the UAV level, lightweight models extract spatial and temporal features from aerial video streams, optimizing resource usage while ensuring privacy preservation and real-time inference. These extracted features are transmitted to a central server, where they are aggregated and refined for robust global event recognition. To enhance adaptability to dynamic UAV environments, the proposed framework incorporates dynamic video chunking and adaptive pooling techniques, effectively handling variable-length video sequences while maintaining temporal coherence. Experimental evaluations on benchmark datasets demonstrate that the SFL framework outperforms state-of-the-art FL methods in terms of event recognition accuracy, computational efficiency, and scalability, making it a viable solution for real-time, low-cost UAV applications in RS and environmental monitoring. This work establishes a scalable, communication-efficient, and privacy-preserving foundation for FL in UAV-based aerial video analysis. Future research will focus on extending the SFL framework to support multimodal data integration, incorporating LiDAR, thermal imaging, and geospatial metadata to enhance event recognition accuracy. Additionally, further optimizations will aim to improve energy efficiency and real-world deployment robustness in diverse UAV-based geospatial monitoring scenarios.

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